Distribution Pump Station Facility-Class Design Basis

A Distribution-Governing Facility Class of the Water Sector Design Basis

DBA-W-FC2 — Facility-Class Design Basis, DBA-W Series

Issued Under: DBA-W-Sector_Framework v1.18; DBA-MA Parent Framework

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1. Purpose and Scope

1.1 Purpose

This document establishes the facility-class design basis for the distribution pump station — the facility class that governs *delivered pressure* across the distribution system, and the second instrument developed in the DBA-W series. At the resolution of the pump-station asset class and site-neutrally, it develops the bounding events, consequence chains, design parameters, adequacy result, and dual design-basis architecture that the sector framework establishes in general terms but does not develop for this class. It returns the same three products as FC1: a facility-class design envelope, a military-action-extension characterization, and an inheritance map. Where FC1 governs the production of safe water, FC2 governs its delivery under pressure to the tap.

1.2 Scope

The class is the distribution pump station in its principal forms — in-line booster stations, high-service stations that lift finished water into the distribution system, and pumped-storage stations that fill elevated storage — together with the pressure-control, surge-protection, and SCADA assets that regulate them. The boundary with FC1 is drawn at the treatment plant's finished-water discharge: raw-water intake pumping and in-plant finished-water pumping — together with clearwells and on-plant storage used to achieve disinfection contact time, which are part of the treatment barrier — are treated as part of the treatment-plant facility class (FC1); pumping that moves finished water through the distribution network and its pressure zones is FC2. This keeps the two classes on the framework's source-to-tap pathway without overlap. Storage-reservoir integrity (FC3) and wastewater (FC5) are out of scope except where a pump station's boundary conditions depend on them.

1.3 The Reference Stations

Two reference configurations bound the class, the pump-station analog of FC1's reference plants. The large high-service or booster station serving a major pressure zone is the governing configuration on consequence: its loss depressurizes a large served population and the critical facilities within the zone. The small skid-mounted or packaged booster serving a single pressure district is the governing configuration on fleet exposure: it is the numerous, lightly staffed, remotely operated case that defines the population of exposed sites. Requirements stated for the class are validated against the large station; the small station governs where the concern is the number of sites an adversary or a common-mode failure can reach at once.

1.4 Relationship to Governing Documents

This instrument is issued under DBA-W-Sector_Framework v1.18 and inherits the DBA-MA parent framework. It does not restate the sector threat taxonomy, bounding methodology, event categories, adequacy standards, or command-broker and Bright Line architecture; it applies them at pump-station resolution and points back to the framework (Section 8 is the inheritance map).

Section references of the form "framework §N" resolve against DBA-W-Sector_Framework v1.18.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application, and the obligation it creates — is defined in FD-BL (Foundational Definition — The Bright Line), which governs. The Broker role that discharges that obligation, the Line-to-Broker pairing invariant, and the competence rule are defined in FD-BR. Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD. Two determinations bind under FD-BL: FD-BL-D1 governs the mode by which the authorization obligation may be discharged, and FD-BL-D2 governs the integrity of the basis on which an authorization is granted. FD-EV reserves the term “envelope”. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it, and adopts the water-sector conservative implementation stated in DBA-W-Sector_Framework §5.5. Instruments of the Foundational Definitions family are cited by designation and without a version token, under the citation-form determination; FD-REG, the Governing Stack Register, is authoritative for their version state and resolves each designation to the issue in force.

Handling. The sections of this document are not equally sensitive, and they are not marked separately in the text. Some set out design parameters and dependencies in enough detail that I share them only with defenders and regulators. Others, such as the statement of what the facility must be built to withstand, can be read by anyone. The Design Basis Advisor, which answers questions across this body of work, keeps a marking for each section and leaves a section out of its answer when that section is marked above what the person asking may see. A reader who has been given this document whole has been given every section.

2. The Distribution Pump Station Facility Class

2.1 Defining Characteristics

For design-basis purposes the distribution pump station is defined by three features. First, a single governing function: maintaining distribution pressure within a band bounded below by the pressure required to sustain service and prevent intrusion, and above by the pressure the mains can withstand without rupture. Second, its role as a network node, not a treatment barrier: the pump station adds no water quality, so its consequence is expressed entirely through pressure and flow, and it propagates rather than contains. Third, near-total dependence on two external systems — the electric grid that drives the pumps and the remote and automated control that regulates them — so that the pump station has no autonomous operating mode of its own and inherits the reliability of the systems it depends on. A single governing function expressed as pressure, a network-node rather than barrier role, and dependence on external power and control: these three define the class and distinguish it from the treatment plant, whose consequence is expressed through chemistry and whose barriers are local.

2.2 The Governing Configuration

The large high-service or booster station on a critical pressure zone is the governing configuration, because it is the bounding case on both consequence chains developed in Section 3: it serves the largest population, holds the largest flow and pressure authority, and typically carries the most automated and remotely reachable control surface. It governs over the pumped-storage alternative, whose consequence is buffered by the storage it fills and is therefore better treated as a boundary interaction with FC3 than as the bounding pump-station case. Requirements stated for the class are validated against the large high-service or booster configuration.

2.3 Deployment Typologies and Size Range

The class spans large regional booster stations, high-service stations at the head of the distribution system, pressure-zone transfer stations, and small district skids, across in-line boosting, high-service lift, and pumped-storage roles, and across the full size range from stations serving hundreds of thousands to single-district skids. A distinguishing property against FC1 is that pump stations are *numerous and distributed*: a single utility operates many of them, most unstaffed and remotely controlled. The class is therefore defined as much by fleet exposure — the number of stations a common-mode or coordinated event can reach at once — as by the consequence of any single site. This fleet property is what makes the coordinated cyber scenario of Section 5.4 the class's distinctive military-action case.

2.4 Consequence Context

Pump-station outage is the governing *pressure* consequence event of the distribution system, the pump-station counterpart to FC1's governing loss of disinfection. Loss of pressure interrupts supply on the timescale set by the pressure zone's storage buffer and demand, and — below a protective threshold — converts a hydraulic event into a public-health event by admitting intrusion into the distribution main. Sustained positive pressure is also the condition under which fire-suppression flow is available, so a pump-station consequence couples directly to the life-safety and fire dependencies the framework carries. The class inherits the framework's mapping of the critical functions to federal Emergency Support Function activation thresholds (framework §4.3); a loss of pressure to a served population or to critical facilities within a zone is an ESF-activating event.

3. The Facility-Class Design Envelope

3.1 Why the Pump Station Is a Distribution-Governing Class

The framework organizes the sector along the source-through-treatment-through-distribution-to-tap pathway (framework §10.2). The pump station governs the distribution leg because sustained pressure is the condition under which treated water reaches the tap safely: treatment can be perfect and the water still be unsafe or unavailable if pressure is lost between the plant and the tap. FC1 governs whether the water is safe to deliver; FC2 governs whether it is delivered safely.

This instrument develops the envelope that responsibility implies rather than re-deriving the placement, which the framework already fixes.

3.2 The Two Consequence Chains

The pump-station class is distinguished, as FC1 is, by the fact that a single initiating event can open two independent consequence chains.

Chain P — loss of pressure and supply. Manipulation, mechanical failure, control-loop failure, or loss of power drops delivered pressure below the level that sustains service and firefighting flow. The chain runs through the distribution hydraulics to the served population as a loss of water and of fire-suppression capability, on the timescale of the zone's storage buffer and demand.

Chain I — intrusion and contamination. When pressure falls below the protective floor — and acutely when a transient drives pressure to zero or negative — the low-pressure condition draws contamination into the distribution main by two distinct mechanisms. The dominant pressure-driven mechanism is *infiltration*: groundwater and its contaminants entering through leaking joints, cracks, and submerged defects in poor-condition pipe that, at normal positive pressure, would leak outward rather than admit inflow. A separate mechanism is *backflow* through a cross-connection, which is primarily the failure or misconfiguration of a backflow preventer or air gap rather than a pressure event as such, though low pressure removes the margin that would otherwise hold it in check. The chain runs through the water to the population as a public-health event, and it does not require the water to have been unsafe when it left the plant.

The two-chain framing is adopted as the analytical heart of the class, and Chain P is treated as triggering Chain I once delivered pressure falls below the protective floor. The protective floor is the minimum positive pressure the distribution system must maintain under all conditions — conventionally the state-mandated distribution-system minimum (commonly on the order of 20 psi, with normal operating pressure well above it) — with the acute intrusion condition being the negative or column-separation pressure that a hydraulic transient can produce. The precise floor and the transient threshold are quantitative values reserved to Section 9. This pressure-to-public-health coupling is the FC2 analog of FC1's disinfection-to-chemical-release pair.

3.3 Design Parameters at Class Resolution

Four parameters govern the envelope. The minimum delivered pressure and its protective floor set where Chain P triggers Chain I; the class parameter is the margin the station must hold above the floor and the demand and fire-flow conditions under which it must hold it. The time-to-depressurization on pump trip sets how long the zone has before a loss of pumping becomes a loss of pressure; the class parameter is that interval as a function of the zone's storage buffer and demand, and it is the recovery-time boundary condition against FC3. The hydraulic transient ceiling sets the surge the station must ride through without rupturing a main or driving pressure negative; the class parameter is that ceiling and the surge-protection standard that bounds it. The pump-capability margin bounds the upper band: the pressure the pumps can reach at full speed,

against what the weakest main in the zone can carry in its present condition; the class parameter is that margin, and it governs the sustained overpressure event of Section 4.8. This instrument states the parameters and their governing logic; their quantitative values are pending validation with utility operators and state primacy agencies (Section 9), consistent with the development sequence the framework specifies (framework §10.4).

4. Design Basis Events: Non-Adversarial Envelope

The following event categories are the facility-class instantiation of the framework's general event categories (framework §4.2), together with the categories specific to a pumped, pressure-governing node — the hydraulic transient and the low-pressure intrusion mechanism — that the treatment-oriented sector categories do not provide. Each is stated in the framework's structure: **the postulated event, its consequence threshold, its required function, and its design-basis duration** — with the consequence chain or chains (Chain P, Chain I) identified in the threshold, and, in the cyber case, the empirical reference events also cited. Requirements the framework already establishes are inherited, not repeated.

4.1 Loss of Pressure / Hydraulic Capacity

Postulated event: pump trip, control-loop failure, mechanical failure, or demand exceedance drops the zone below service pressure.

Consequence threshold: delivered pressure below the level that sustains service and firefighting flow (Chain P); and, once pressure falls below the protective floor, intrusion into the distribution main (Chain I). The class parameter stressed is the minimum delivered pressure and its margin above the protective floor.

Required function: the station must sustain, or fail in a manner that sustains, positive pressure above the protective floor for the class duration, or the zone's storage must carry it; a control-loop failure must not be able to command pressure below the floor without a positive local limit.

Design-basis duration: 30 days, the class default (framework §4.2.1).

4.2 Hydraulic Transient / Water Hammer

Postulated event: a sudden pump start or stop — most commonly a pump-power failure — or a rapid valve action generates a pressure surge, driving a downsurge that can reach negative pressure and column separation and an upsurge that can rupture mains or damage the station.

Consequence threshold: an upsurge beyond the transient ceiling that ruptures a main or damages the station (Chain P plus physical damage), or a negative-pressure downsurge, which is the most direct intrusion condition the class carries (Chain I, acute). The class parameter stressed is the transient ceiling and the surge-protection standard.

Required function: surge protection — surge relief, air-vacuum valves, controlled pump ramp, and check-valve slam mitigation — sized to hold the transient within the ceiling and to prevent negative pressure.

Design-basis duration: not duration-governed; the event is a transient pressure wave rather than a sustained condition.

Surge is treated here as its own design-basis event rather than folded into 4.1, because its physics (an inertial pressure wave), its protective measures (surge relief, air-vacuum valves, controlled ramp, check-valve slam mitigation), and its acute intrusion mode are distinct from a steady loss of pressure and are specific to pumping. It is also distinct from sustained overpressure (Section 4.8), whose mechanism is pumps driven up and held rather than a pressure wave, and whose protection is a bound on the pressure the pumps can reach rather than on how fast flow can change. Protection against one does not protect against the other.

4.3 Cyber Attack on Pump-Control Operational Technology

Postulated event: intrusion into the pump-control system manipulates variable-frequency-drive or pump setpoints, pressure controllers, or SCADA logic, or falsifies the pressure indication operators rely on, driving delivered pressure outside the safe band, below it or above it, or producing a transient beyond the class ceiling.

Consequence threshold: pressure or surge driven outside the safe band by manipulated control or falsified indication (Chain P and Chain I).

Required function: the pump-control surface inherits framework §4.2.2 and §5.2 as revised in v1.6 — interlock independence, credential architecture, monitoring, indication integrity, and protected engineering objects — applied specifically to the VFD and pump-start control, the pressure-control setpoint, the pressure-indication path, and the engineering workstation, vendor software, and controller project file for the station. The class's protected engineering objects are the pump-sequencing and pressure-control logic. Against a hydraulic consequence produced by remote command, the class relies on the field protection of Sections 4.1, 4.2 and 4.8, which the network cannot reach.

Design-basis duration: indefinite (framework §4.2.2).

Reference events: this category is directly exercised by three reference events. DBA-W-RE-1 (the multi-site Minnesota campaign) and DBA-W-RE-2 (the single-site Oldsmar manipulation) supply the indication-integrity and attribution-independence findings the class inherits. DBA-W-RE-3 (the hacktivist campaign against small water systems), proposed through this class (Section 8.3), shows that a remote command can produce a hydraulic consequence directly: at its governing incident one command took pressure to zero and a second drove every pump to full speed and held it there until distribution pipe failed (Sections 4.1 and 4.8).

4.4 Loss of Electric Power to Pumping

Postulated event: loss of grid power or failure of automatic transfer stops the pumps; given the class's near-total power dependence this is the dominant common-cause initiator, and the power-failure trip is also the most common initiator of the 4.2 transient.

Consequence threshold: loss of pumping that drops delivered pressure below the service level (Chain P) and, below the protective floor, into intrusion (Chain I).

Required function: on-site standby power, an automatic-transfer path that is itself hardened, or storage sufficient to hold pressure for the class duration, together with surge protection that operates on the power-loss trip without external power (Section 6.2).

Design-basis duration: 30 days, the class default (framework §4.2.1).

4.5 Backflow / Intrusion Under Low Pressure

Postulated event: delivered pressure falls below the protective floor, or negative pressure occurs during a transient, and contamination enters the distribution main — chiefly by infiltration through submerged defects in poor-condition pipe, and secondarily by backflow through a cross-connection whose backflow preventer or air gap has failed or been defeated.

Consequence threshold: any excursion below the protective floor, or any negative-pressure transient (Chain I).

Required function: the station and zone must be designed so that no design-basis pressure event drives pressure below the protective floor without a defined protective response — pressure sustainment, zone isolation, cross-connection control and backflow-preventer integrity, or a boil-water fallback where the floor cannot be assured.

Design-basis duration: not duration-governed; the threshold is any sub-floor or negative-pressure excursion.

This is treated not as an independent initiator but as the cross-cutting consequence limb that links Chain P to Chain I — the mechanism by which the pressure events of 4.1, 4.2, 4.3, and 4.4 escalate into a public-health event. It is stated as its own subsection because it is the pump-station analog of FC1's chemical-release chain: the point at which a hydraulic failure becomes a contamination event, and the reason the protective floor is a design-basis parameter rather than an operating preference.

4.6 Physical Attack on Critical Nodes

Postulated event: destruction of pumps, the control room, or the power feed at a station serving a critical pressure zone.

Consequence threshold: loss of the station's pumping or control at a critical-zone site (Chain P plus a containment-and-restoration burden).

Required function: supply that is robust, diverse and independent — two or more trains, differing in kind, and sharing no location, power supply, control system, vendor or update path, so that no single act defeats them all (framework §4.1), with rapid restoration beside it, for stations whose single-point loss depressurizes a critical zone — redundancy alone, or diverse pumps that share a station, not being credited against deliberate attack (DBA-MA-DiD §3.4) —

and hardening of the power feed and control room commensurate with the station's applicability determination (Section 5.1).

Design-basis duration: 30 days for full restoration; 24 hours for minimum service to critical dependent facilities (framework §4.2.3).

4.7 Natural-Hazard Exceedance

Postulated event: a flood, seismic, or extreme-temperature event beyond design disables the station or its power feed; flooding of a below-grade station and loss of the power feed are the governing modes.

Consequence threshold: loss of the station or its power feed beyond the facility and power design basis (Chain P).

Required function: inherited from framework §4.2.6, with the class addition that surge protection and the manual pressure-control path must remain available under the exceedance event.

Design-basis duration: 30 days for full restoration; 72 hours for minimum service to critical dependent facilities (framework §4.2.6).

4.8 Sustained Overpressure

Postulated event: pumps are driven to full speed, or to a speed above what the zone requires, and held there, by a control-loop or drive failure or by remote command, so that delivered pressure rises and stays above what the weakest main in the zone can carry.

Consequence threshold: rupture of distribution mains where the network is weakest for its pressure, and the loss of pressure that follows (Chain P plus physical damage); once pressure in the ruptured zone falls below the protective floor, intrusion (Chain I). The class parameter stressed is the pump-capability margin of Section 3.3.

Required function: delivered pressure is bounded at the equipment, by protection that the networked control system cannot reach or change except through a governed act (Section 6.2). The preferred protection is the pump-capability bound: pumps whose pressure at full speed stays below what the weakest main in the zone can carry as it is today, so that no command can break it. Where that bound does not hold, the station carries a speed limit set in the drive, a high-pressure trip at the pump, and pressure relief sized for steady flow rather than for a transient. Because the weakest main sets the bound, the high-pressure protection is set against the present condition of the mains, not their design rating, and is reviewed when that condition changes. A limit that the operator interface can override is not a limit against a commanded overpressure.

Design-basis duration: not duration-governed in the transient sense; the condition lasts for as long as the command or the failure holds, and the required function must hold for as long as it does.

Sustained overpressure is treated as its own design-basis event, distinct from the hydraulic transient of Section 4.2. Its mechanism is pumps driven up and held, not a pressure wave; where the damage falls is decided by the network, at its weakest main for the pressure, not by the device that starts the event; and it is bounded by the pressure the pumps can reach, not by how fast valves close or pumps stop. Protection against one does not protect against the other. The event is taken in on the third finding and third recommendation of DBA-W-RE-3 (Section 8.3).

Table 4-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Inherits / adds
Loss of pressure / capacity	Trip, control-loop failure, or demand exceedance below service pressure	Chain P (→ I below floor)	framework §4.2
Hydraulic transient / surge	Sudden start/stop or valve action; down- and up-surge	Chain P + damage; Chain I (acute)	adds (class-specific)
Cyber attack on pump OT	VFD/setpoint/pressure-control manipulation; falsified indication	Chain P / I	framework §4.2.2, §5.2 (v1.6)
Loss of electric power	Grid loss or transfer failure stops pumping	Chain P / I	framework §4.2.1
Backflow / intrusion	Pressure below protective floor or negative transient	Chain I (limb of P)	adds (consequence mechanism)
Physical attack on nodes	Strike on pumps, control room, or power feed	Chain P + restoration	framework §4.2.3
Natural-hazard exceedance	Hazard beyond facility/power design	Chain P	framework §4.2.6
Sustained overpressure	Pumps driven up and held above the weakest main’s capacity	Chain P + damage; Chain I below floor	adds (class-specific)

4A. The Adversarial Design Basis for the Class

The framework's three-tier threat-actor taxonomy (framework §7.5.1) assigns to this class as follows. Tier 3, the sophisticated non-state actor, is the class's non-adversarial cyber baseline — the Oldsmar and Minnesota reference events sit here — and is addressed by the Section 4 requirements. Tier 1 (state) and Tier 2 (near-state proxy) trigger the military-action extension of Section 5. The class adopts the v1.6 refinement that tier denotes direction, not a ceiling on capability, and it carries a distinctive amplifier the treatment class does not: because pump stations are a numerous, uniform, remotely controlled fleet, a single understood control architecture can be applied against many stations at once, so the class assesses capability against the fleet, not only the single site. The governing adversarial event for the class is the coordinated manipulation of pump control across multiple stations to collapse pressure system-wide; under the military-action extension it is joined by the precision strike on a critical-zone station.

5. Military Action Extension

5.1 MA Extension Applicability Determination

Every pump station performs the facility-level military-action applicability determination the framework makes mandatory (framework §7.1): the primary factor is geolocation relative to military-action exposure, and the secondary factor is strategic value — here the population and the critical facilities of the pressure zone the station serves, and the station's position in the fleet. For this class the determination has a fleet dimension absent from FC1: a station may carry the extension not only for its own served population but because it is one of a set whose coordinated loss is strategically significant. The determination is time-varying, recorded, and re-evaluated on the framework's triggers, and any scope reduction carries a recorded basis (Section 7).

5.2 The Four Characteristics Applied to the Class

Applied to the pump station, the framework's four military-action characteristics (framework §7.3) take the form below, and the class requirements are written against each. Intentionality: a station is chosen for the consequence of the zone it serves, so the class requirements are set by that consequence and not by the likelihood of its selection. Persistence: an adversary may hold access to pump control over a long interval before acting, so the protective functions of Section 6.2 must not depend on the networked control system being uncompromised. Precision: an adversary may act on the specific function whose defeat produces the governing consequence, so each protective function must hold on its own and not on the assumption that another has held. Denial: an adversary may deny the operator accurate indication and the margin to respond, so the class carries the framework's indication-integrity requirement and an adequacy condition that holds without external communications (Section 7.1). The class validation standard is the state actor of the framework's taxonomy (framework §7.5.1), assessed against the fleet as well as the single site (Section 4A).

5.3 The Governing MA Scenario

The governing military-action scenario for the class is a precision strike or sabotage on a station serving a critical pressure zone, under conditions chosen to make restoration as hard as possible. It realizes Chain P at the zone scale and, through the ensuing depressurization, Chain I. The extension therefore requires hardening and standoff for critical-zone stations commensurate with their applicability determination, and a restoration path, by mobile pumping, zone isolation or storage transfer, that survives the strike and does not assume normal demand, staffing or grid supply. This is the FC2 analog of FC1's chemical-storage strike: the single action that opens both chains.

5.4 The Coordinated Cyber Scenario

The governing military-action cyber scenario is the simultaneous manipulation of pump control across many stations, so that delivered pressure is driven outside the safe band across the system at once. It is directly informed by the reference events. DBA-W-RE-1 established that a

coordinated multi-site campaign against a common device and configuration population is a realized event class. DBA-W-RE-2 established the single-site setpoint-manipulation mechanism and the attribution-independent reading of it. DBA-W-RE-3 showed remote commands producing hydraulic consequences across a campaign against small systems. Against this scenario the class relies on the common-substrate independence test and the indication-integrity and protected-engineering-objects requirements the framework carries (framework §4.1, §5.2), and on field protection at each station that holds the safe band whatever the network commands (Sections 4.8 and 6.2). A control architecture shared across a fleet is a common substrate, and the requirement is that no single action through it can defeat the protection of every station it reaches.

5.5 Cross-Sector Interaction

The pump station is the water sector's most acute cross-sector coupling point. Its near-total dependence on electric power makes the water-electricity interdependency (framework §8.1) both an independent event category (4.4) and a force multiplier for every other scenario, and its dependence on communications and SCADA couples it to the information-technology and communications sectors for its control. Where a pressure event is timed to a concurrent loss of grid power, the energy-side consequence and restoration are addressed by the DBA-EN energy work; this instrument holds the water-side consequence and names the coupling. Where this station is the sole cooling-water supply path for a thermoelectric generation facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-ES series, that supply relationship carries tightened acceptance criteria, which discharges for this class the reverse-direction obligation DBA-EN-SF §5.4 places on the DBA-W series (framework §8.1). The criteria are quantified with the operator of the generation facility; who bears the cost of meeting them is open and is not settled by this instrument.

6. The Two Design Bases

6.1 The Two Design Bases

The pump station carries two design bases simultaneously (framework §10.1): a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here. The two meet at the point developed in Section 6.4.

6.2 The Physical-DB Half: The Determinism Posture

The physical design basis is the set of protective functions the station relies on to hold its safe state independently of the networked control system: local pressure protection that limits delivered pressure to the safe band, including a speed limit set in the drive and a high-pressure trip at the pump and, where the station’s pumps allow it, the pump-capability bound of Section 4.8; surge protection that operates on a pump-power-loss trip without external power or SCADA, and a manual or hard-wired trip and pressure-control path independent of the networked control

system. Those among them that decide on measured quantities against setpoints are to be locally sovereign as FD-LS §3.2 defines the property: the decision is physical law, the function is colocated with the station, and no authority is reachable in either limb, so that nothing on the reachable side can withhold, delay or induce its action, and nothing there can alter its setpoints, logic or bypass state except through a governed act. FD-LS governs and is not restated here. Above the Bright Line — the boundary drawn on the consequence of each individual action — a component within scope of FD-LD §4.3 may propose but may not execute without command-broker authorization. The determinism posture is a scope posture, not a placement. The pressure-limit and surge-protection functions, where deterministic within a stated input bound and formally testable as deployed, meet the carve-out and are outside the Command Broker obligation for their own in-bound actions (FD-LD §4.3). They are not below the line: they take some of the highest-consequence actions at the station, and they are outside the obligation because their behavior is bounded by proof (FD-BL §4.5). Any pressure-management component that cannot be shown to meet the carve-out is within scope; each of its actions is placed on its own consequence (FD-BL §4.2), and under the conservative implementation of framework §5.5 each action it would carry into the physical command layer is treated as above the line unless a documented per-action assessment places it below. A change an in-scope component proposes to a protective function’s setpoints, logic, bypass state or actuation criteria is an action upon that function, placed on its own consequence (FD-BL §4.4).

6.3 The AI-Governance-DB Half: Named and Referenced

Machine-learning and generative-AI systems in pump operation — pressure and pump-sequencing optimization, demand forecasting, and autonomous pressure management — are governed by the Command Broker and Bright Line architecture applied at framework §5.5 and established cross-sector in the Command Broker family (CB-Framework_Cross_Sector; CB-REG, the Command Broker Family Register, is authoritative for its version state). This instrument does not redefine that architecture; it names it and assigns to it the station's highest-criticality action, a change to a pressure setpoint or a pump-sequencing command that can move the zone toward the protective floor or a transient. Placement is by the consequence of each action, not by the component (FD-BL §4.2, §4.3). A forecasting or optimization model that only proposes is within scope unless it meets the carve-out (FD-LD §4.3), and each change it proposes, together with an operator’s acceptance of it, is one action placed on the consequence of that change; its output that would make no change of its own is placed below the line and carries the indication-integrity obligation of FD-BL-D2 wherever it can ground an authorization. An in-scope autonomous controller empowered to execute a setpoint change on a critical zone takes actions whose consequence places them above the line, and each requires command-broker authorization.

6.4 The Interaction Point: Control Authority Under Denial

The two design bases meet at the retention of safe pressure control when SCADA and automation are denied or manipulated — the pump-station analog of FC1's autonomous-safe-

condition-under-denial. The control system must be architecturally incapable of executing an above-line action originating from a component within scope of FD-LD §4.3 — including any pressure or pump command that can drive the zone below the protective floor or into a design-basis transient — without command-broker authorization, and the physical pressure and surge protection of Section 6.2 must hold the safe band with no reachable authority of any kind. The physical design basis guarantees the safe band; the AI-governance design basis guarantees that no proposing system can move the station out of it without an enforceable human authorization.

6.5 Scope Boundary

This instrument covers the pumping and pressure-control node. It does not cover treatment chemistry, which is FC1; storage integrity and the storage buffer, which are FC3; or wastewater pumping and conveyance, which are FC5. Where a pump station's consequence depends on those classes — most directly the storage buffer that sets its time-to-depressurization — this instrument names the dependency and defers the owning analysis to the adjacent class.

7. Design Basis Adequacy and Determination

This section states whether the class-resolution design basis is adequate and what determination it supports; it is the pump-station development of FC1's adequacy result and carries the mandatory consequence-basis conditionality check.

7.1 The Autonomous Safe Pressure Condition

The class adequacy result is the Autonomous Safe Pressure Condition: the condition in which, on loss of networked control and external support, the station and its zone hold delivered pressure within the safe band autonomously. Pumps fail in a manner that does not drive a damaging transient; surge protection actuates passively on the power-loss trip; local pressure limits, set against the present condition of the weakest main, hold the upper band whatever the network commands; a manual pressure-control and zone-isolation path is operable without SCADA; and, where the protective floor cannot be assured, a boil-water fallback preserves the public-health boundary. The condition must be achievable and sustainable by a minimal on-site or mobile crew for a defined duration without external communications, inheriting the local-sovereignty principle of framework §9.4.

7.2 Adequacy in the Non-Adversarial and Bounded-Adversarial Domains

In the non-adversarial domain the Autonomous Safe Pressure Condition is achievable with the Section 4 requirements: fail-safe pump control, passive surge protection, standby power or storage for the class duration, and manual pressure-control procedures maintained by trained operators. In the adversarial domain adequacy is bounded rather than unconditional: against the cyber chain, indication integrity, protected engineering objects, and the common-substrate independence test bound the event and preserve the safe pressure band even when displayed values are falsified; against the physical chain, hardening, standoff, independent protection of at

least one element of every governing cutset (DBA-MA-DiD §3.4), and a surviving restoration path bound it. Adequacy holds within these bounds.

7.3 The Declared Residual

Two residuals are declared rather than concealed. First, a sufficiently large or well-timed precision strike on a critical-zone station can realize Chain P at the zone scale regardless of cyber posture; hardening and standoff reduce but do not eliminate it, and beyond the design strike it is a declared residual driving independent restoration paths and emergency planning. Second, the fleet property means a coordinated action against a shared control architecture can exceed the restoration capacity of a single utility; beyond that capacity the residual is a system-wide loss of pressure that mutual aid and the recovery architecture, not the single station's autonomy, must address. These residuals set the boundary of the class adequacy claim. They are the declared residuals of the DBA-MA-DiD §3.5 adequacy statement, which framework §4.1 makes owed.

7.4 Regulatory and Institutional Basis

The class's protective floor and its adequacy determination rest on the Safe Drinking Water Act and the state primacy agencies that set and enforce the distribution-system minimum-pressure requirement, on America's Water Infrastructure Act risk-and-resilience and emergency-response obligations, and on the American Water Works Association pressure and surge-control practice that supplies the engineering standard for the transient ceiling of Section 4.2. Engineering practice is the near-term propagation vehicle for the class requirements ahead of any mandate (framework §2.5). The quantitative floor, margin, and surge standard are set against these authorities and remain open pending validation (Section 9).

7.5 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter — a reduced pressure margin, a reduced standby-power duration, a reduced surge standard, or a reduced redundancy for a critical-zone station — must be justified against the class baseline under the consequence-basis conditionality principle the framework makes mandatory for every facility-class instrument. The scaling corollary applies: the larger the claimed relaxation relative to the conservative baseline, the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis.

8. Inheritance

8.1 The Inheritance Map

The framework requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally (framework §10.1). Table 8-1 is that map for this class.

Table 8-1. Inheritance map for DBA-W-FC2.

Element	Source in framework v1.18	Inheritance mode
Threat-actor taxonomy	§7.5.1	Inherited unchanged
Design-basis principles / bounding method	§4.1	Inherited unchanged
Cyber requirements (interlock, credential, monitoring, indication integrity, protected engineering objects)	§5.2	Inherited unchanged, applied to the pump-control surface
Command Broker / Bright Line architecture	§5.5; Command Broker family (CB-REG)	Inherited unchanged, referenced
Common-substrate independence test	§4.1 (v1.6)	Inherited unchanged; central to the fleet scenario (§5.4)
Event categories	§4.2	Inherited with adaptation; adds hydraulic transient, sustained overpressure, and the backflow-intrusion mechanism
MA-applicability determination	§7.1	Inherited with adaptation to facility and fleet level
Adequacy standard / local sovereignty	§9.4	Inherited with adaptation as the Autonomous Safe Pressure Condition
Consequence-basis conditionality check	framework portfolio requirement	Inherited with adaptation
The two consequence chains (pressure / intrusion)	§10.2 (scope)	Locally developed
Class design parameters and the ASPC	—	Locally defined
Two design bases (physical and AI-governance)	§10.1	Inherited unchanged; stated in full at Section 6

8.2 Anticipated Derivatives

Site-, state-, and contract-specific pump-station design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. Two derivative instruments were anticipated under this class. The coordinated-pressure-collapse use case, tracing the fleet cyber scenario of Section 5.4 against a worked pressure-zone network, is issued as DBA-W-UC-2 (Coordinated Pressure Collapse Across a Distribution Network). The pump-station power-dependence companion, developing the water-electricity coupling of Section 5.5 jointly with the DBA-EN energy work, remains anticipated.

8.3 Reference Events Proposed Through This Class

DBA-W-RE-3 (Hacktivist Attacks on Small Water Systems), on the pro-Russia hacktivist campaign against small water systems from January 2024 to summer 2026, is proposed through this class and conferred at framework §10.5 on the author's ruling of 9 November 2026. The instrument was issued ahead of the conferral, on the author's direction. Its governing incident, at

Tureby in Denmark, was sustained overpressure by command and not surge, and this class carries it at Section 4.8 from this version. Of the instrument's recommendations, R-3 disposes to this class for conformance and is taken up here: sustained overpressure and water hammer are distinct events with distinct field protection (Sections 4.2 and 4.8), the preferred protection against overpressure is the pump-capability bound, and high-pressure protection is set against the present condition of the mains. The findings the instrument returns to the framework are taken up at the framework's next conformance pass.

9. Open Items

9.1 Quantitative Design Parameters — Pending Operator and State-Primacy Validation

The minimum delivered pressure, the protective floor, the margin above it, and the time-to-depressurization named in Sections 3.2 and 3.3 are stated as parameters, not numbers. Per framework §10.4, FC2 development requires engagement with utility operators and state primacy agencies to validate them against real pressure-zone architectures; the quantitative values await that engagement. These parameters are now anchored to a published design standard against which that engagement is a confirmation rather than an origination: the Recommended Standards for Water Works (Ten States Standards, 2022) set a minimum distribution pressure of 20 psi (140 kPa) at all points under all conditions of flow, including fire flow, and a normal working band of 60 to 80 psi that shall not fall below 35 psi (§8.2.1), and require booster pumps to hold at least 20 psi in the suction line under automatic low-pressure control (§6.4.1) — the published protective-floor value for Chain P and the published intrusion-floor value for Chain I. A standby or auxiliary power source for continuous service on loss of primary power is required by §6.3.4. Because many state primacy agencies adopt the Ten States Standards by reference, the remaining operator engagement confirms the local floor and the margin the station must hold above it rather than establishing the floor itself. The section-by-section mapping is recorded in the DBA-W Ten States Standards design-value crosswalk (v0.1).

9.2 Transient / Surge Standards

The numerical transient ceiling and the required surge-protection standard for Section 4.2 are open and must be set against American Water Works Association surge practice and the station's specific hydraulics. This is the most class-specific open item and has no counterpart in FC1. The Ten States Standards close the requirement but not the number: §6.6 requires surge-relief valves or slow-acting check valves to minimize hydraulic transients, yet sets no numeric surge ceiling, so the ceiling remains a site surge analysis against AWWA practice as stated.

9.3 Sub-Threshold Minimum-Requirement Set

The bounded minimum-requirement set for small stations below the applicability threshold (Section 2.3) is a forthcoming instrument, linked to the framework §1 sub-threshold scope

statement and to reference event DBA-W-RE-1. Its relationship to this class — which requirements bind the small district skid and which are relaxed — is open.

9.4 Groundwater / Small-System Variant Scope Reduction

The specific class requirements that fall away for small or simple stations, the pump-station analog of FC1's groundwater-variant reduction, are recorded in principle in the inheritance map but require validation against representative small-station architectures before the reduction is fixed.

9.5 Facility-Class Use Case

The first postulated-scenario use case under FC2, the coordinated-pressure-collapse use case of Section 8.2, is issued as DBA-W-UC-2 (Coordinated Pressure Collapse Across a Distribution Network). It exercises both consequence chains against a worked fleet scenario. It was issued under an earlier version of this class, and its conformance to this version, including the sustained overpressure event of Section 4.8, is open for its next revision.

9.6 The Upper-Band Bound Against the Condition of the Mains

The pump-capability bound and the high-pressure setpoints of Section 4.8 are set against the present condition of the weakest main in each zone, which only the utility's own condition records establish. The class states the bound as a parameter. Its value, and how often it is reviewed as mains age or are replaced, are open pending validation with utility operators and state primacy agencies, with the other quantitative parameters of Section 9.1.

Boundary note carried from the register: FC2 sits between FC1 (treatment) and FC3 (storage) on the source-to-tap pathway. The handoffs are the treatment plant's finished-water discharge (with FC1, resolved at Section 1.2) and the storage buffer that sets the pump station's time-to-depressurization (with the forthcoming FC3 storage instrument, named at Sections 3.3 and 6.5). Confirm both when FC3 is drafted.

References

DBA-W Series and Parent Framework

DBA-W-Sector_Framework v1.18. Design Basis for Water Sector Infrastructure. Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued and from which it inherits (Section 8).

DBA-W-RE-1 v0.8 and DBA-W-RE-2 v0.3. Reference Event Instruments (Minnesota community water systems, 2026; Oldsmar water treatment plant, 2021). Eclectic Technologies, 2026. Empirical basis for the cyber event category (Section 4.3) and the coordinated-cyber scenario (Section 5.4).

DBA-W-RE-3 v0.5. Hacktivist Attacks on Small Water Systems. Eclectic Technologies, 2026. Reference event proposed through this class (Section 8.3); empirical basis for the sustained overpressure event (Section 4.8).

DBA-W-FC1 v0.6. Water Treatment Plant Facility-Class Design Basis. Eclectic Technologies, 2026. Adjacent facility class; boundary drawn at the treatment plant's finished-water discharge (Sections 1.2, 6.5).

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics and the design-basis-under-military-action methodology.

Command Broker Framework Series

Command Broker and Bright Line architecture (framework §5.5; CB-Framework_Cross_Sector, cited by designation, version state in CB-REG). Governing instruments for the AI-governance design basis named and referenced at Sections 6.3 and 6.4.

Regulatory and Standards

Safe Drinking Water Act and the state primacy agencies that set and enforce the distribution-system minimum-pressure requirement; America's Water Infrastructure Act (2018) risk-and-resilience and emergency-response obligations; American Water Works Association pressure and surge-control practice (Section 7.4).

Recommended Standards for Water Works (Ten States Standards), 2022 Edition. Great Lakes–Upper Mississippi River Board of State and Provincial Public Health and Environmental Managers. Published design standard supplying the distribution-pressure floor and working band (§8.2.1), the booster-pump suction-line intrusion floor (§6.4.1), the standby-power requirement (§6.3.4), and the surge-mitigation requirement (§6.6) cited at Sections 7.4 and 9. Mapped in the DBA-W Ten States Standards design-value crosswalk v0.1.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	August 2026	Tim Roxey	Initial issue. Second facility-class instrument in the DBA-W series: the distribution pump station. Establishes the two consequence chains — loss of pressure (Chain P) and low-pressure intrusion (Chain I) — and the protective-pressure floor (§3); the facility-class event envelope, adding the hydraulic-transient and backflow-intrusion categories to the inherited set with each event stated on the framework's four fields (§4); the adversarial tier taxonomy with the fleet amplifier (§4A); the military-action extension including the coordinated multi-station cyber scenario and the precision strike on a critical-zone station (§5); the dual design basis with the interaction point under denial (§6); the Autonomous Safe Pressure Condition adequacy result and declared residuals (§7); and the required inheritance map (§8). Inherits DBA-W-Sector_Framework v1.6. Quantitative design parameters (minimum pressure, protective floor, time-to-depressurization,

Version	Date	Author	Description
			surge ceiling) flagged pending operator and state-primacy validation (§9).
0.2 — DRAFT	August 2026	Tim Roxey	West Yost review (A. Ohrt) — clear edits: Chain I intrusion split into its two mechanisms — pressure-driven infiltration through submerged defective pipe (dominant) and backflow through a failed cross-connection / backflow preventer (§3.2, §4.5); cross-connection control added to the §4.5 required function; clearwells and on-plant contact-time storage stated as in-plant (FC1) at the FC1/FC2 boundary (§1.2). Structural framing unchanged.
0.3 — DRAFT	August 2026	Tim Roxey	Quantitative closure of Section 9 against the Recommended Standards for Water Works (Ten States Standards, 2022): the protective-pressure floor and working band anchored to the 20 psi all-conditions minimum and 35–80 psi band (§8.2.1); the Chain I intrusion floor anchored to the 20 psi booster suction-line minimum (§6.4.1); standby power to §6.3.4; and the surge item recorded as requirement-closed but ceiling-open, since §6.6 mandates transient mitigation without a numeric ceiling. Values framed as confirmation, not origination, because state primacy agencies commonly adopt the standard by reference. Ten States Standards added to References. No normative or structural change.
0.4	September 2026	Tim Roxey	FD citation form conformed under the citation-form determination. The governing-instruments paragraph cited the Foundational Definitions with version tokens, two of which had gone stale on 3 September (FD-BL-D1 re-issued at v0.4, FD-BL-D2 at v0.3) — a version-pinned citation resolving into superseded text. Tokens dropped; the family is cited by designation and resolves to FD-REG. No requirement, event category, parameter or finding changed.
0.5	October 2026	Tim Roxey	Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026 (P9-07), together with the acts OI-14 placed at this revision. §6.2 no longer places the protective functions below the line as deterministic. The determinism posture is stated as scope: the functions are outside the Command Broker obligation by the carve-out and not by placement (FD-LD §4.3; FD-BL §4.5), and a component not shown to meet the carve-out is within scope with each action placed on its consequence, the conservative implementation of framework §5.5 keeping the outcome the earlier text reached. Its local-sovereignty sentence is read as FD-LS's construct and cites FD-LS §3.2 with both limbs and the governed act. §6.3 places by the consequence of each action rather than by component (OI-14). §6.4 states the obligation's reach by origin in a component within scope of FD-LD §4.3 rather than as AI-originated (OI-14). Whether each outcome so restated is right on consequence is the author's to confirm. The Command Broker family is cited by designation against CB-REG rather than by project at §6.1, §6.3, the inheritance map and References, a path citation found beyond OI-4, which named FC1 only. The parent framework is cited at v1.13 at the issued-under and parent-document lines, §1.4 and References (OI-3); provenance notes recording a value established at v1.6 stay as written. Reference-event and adjacent-class pins are repaired to current. The required function for single-point-loss

Version	Date	Author	Description
			nodes credits independent protection, with redundancy kept beside it against random loss and not in its place, the adversarial adequacy statement no longer credits redundancy, and the declared residuals of §7.3 are identified as the DBA-MA-DiD §3.5 output that framework §4.1 now makes owed (OI-11). OI-12, the reverse-direction cooling-water rule of DBA-EN-SF §5.4, is not taken at this revision and stays open for the author. Held, not missed: the local-sovereignty principle cited from framework §9.4 at §7.1 was read and left as written, pending Project 5's answer on the DBA-MA construct. Revision-history rows were excluded from substitution.
0.6	October 2026	Tim Roxey	Two author decisions of 3 October 2026. The required function for single-point-loss nodes credits trains only when they are robust, diverse and independent, sharing no location, power supply, control system, vendor or update path, so that no single act defeats them all (framework §4.1); redundancy alone, or diverse trains that share a room, are not credited against deliberate attack. Section 5.5 carries, conditionally, the reverse-direction cooling-water obligation of DBA-EN-SF §5.4: where the facility is the sole cooling-water supply path for a Tier 2 or Tier 3 DBA-ES thermoelectric facility, that supply relationship carries tightened acceptance criteria, quantified with the generation facility's operator, and who bears their cost is left open (OI-12 closed). The parent framework is cited at v1.14 and adjacent-class pins are repaired to current. Revision-history rows were excluded from substitution.
0.7	November 2026	Tim Roxey	On the author's determinations of 10 November 2026 recorded in the return note to Project 10 on making FD-LS and this class citable for vendors, and his decision of 11 November 2026 to take in commanded sustained overpressure. The five confirmation markers at §1.2, §2.2, §3.2, §4.2 and §4.5 are struck, all five calls being confirmed as written on the West Yost review. §4.3, §5.2, §5.3 and §5.4 are rewritten to the TLP:CLEAR baseline: §4.3 no longer names a technique for inducing a transient, §5.2 states each military-action characteristic as the requirement it drives, and §5.3 and §5.4 no longer carry scenario timing or the means by which one action reaches the fleet. A new §4.8 carries sustained overpressure as its own design-basis event, distinct from the transient of §4.2, on the third finding and recommendation of DBA-W-RE-3 v0.5; §3.3 adds the pump-capability margin as a fourth parameter, §6.2 and §7.1 add the field protection that bounds the upper band, Table 4-1 and the inheritance map add the event, and §9.6 registers the bound's value as open. New §8.3 records that this class proposed DBA-W-RE-3, conferred at framework §10.5. §9.5 is corrected: the coordinated-pressure-collapse use case is issued as DBA-W-UC-2, and §8.2 says so. The facility-class handling paragraph is added at the end of §1 under the Determination that use cases and facility classes carry one handling paragraph. The parent framework is cited at v1.18 at the issued-under and parent-document lines, §1.4, the inheritance map and References; framework v1.18 adds the §10.1 statement of the two design bases that §6.1 and §8.1 cite, which the v1.5 framework entry recorded but the body never carried. Reference-event pins are

Version	Date	Author	Description
			repaired to current, and DBA-W-RE-3 is added to References. The cover date is corrected from August to November 2026. Revision-history rows were excluded from substitution.